

MEC 125 NOTE

BRIEF HISTORY OF AUTOMOTIVE INDUSTRY IN NIGERIA

The history of Automobile in Nigeria started earlier before Nigeria as a country had her independence in 1960. The western world introduced the use of automobile to Nigeria when oil was booming in the 70s. Developed countries like America, Germany, France, and Britain etc. through partnership with Nigerian government and issuing of license of operation and control policy, established six Automobile assembly plants in Nigeria. The establishment of the six automobile assembly plant in Nigeria brought quick development to the country. However, these assembly plants could not stand the test of time due to non-adherence to the law that was established by the federal government to assist and protect the local automobile industries, by the government itself.

Since in the year 2000, it has been recorded that there is a decline in the production of Automobile, some of the Automobile industries that were established between 70s and 80s are no longer in operation, as a result of challenges that has befall them. About 75% existing Automobile companies in Nigeria today deal only on car sales, about 80% of Automobile used in the country today are fairly used cars popularly known as Tokumbo, because of the decline in Automobile production in Nigeria.

SOME AUTOMOBILE INDUSTRIES IN NIGERIA

- i. Peugeot Automobile Nigeria limited (PAN) Kaduna in 1975,
 - ii. Volkswagen of Nigeria Limited (VWON) Lagos 1978,
 - iii. Anambra Motor Manufacturing Limited (ANAMMCO) Emene – Enugu (1980);
 - iv. Steyr Nigeria Limited Bauchi,
 - v. National Truck manufacturers (NTM) Kano Fiat Production.
 - vi. Leyland Nigeria Limited (LNL) Ibadan, between 1970 and 1980
- REASONS FOR THE DECLINE IN AUTOMOTIVE INDUSTRIES IN NIGERIA**

i. Poor Policy Implementation:

The establishment of Automobile industries in Nigeria is based on policy. This law that was established by the federal government to assist and protect the local Automobile industries has been pushed aside; by the government itself. The

government import foreign made Automobile rather than patronizing Nigerian made Automobile cars.

ii. Poor Leadership

In the past 20 years the government of Nigeria is not interested in the development of technological skills among her citizen. Every government believes that with the country natural resource which is the crude oil. The country stand not to face any economic crisis hence many past government leaders of this nation has mismanage the fund of the nation by siphoning money abroad building houses and companies abroad rather than investing money on local industries that will help to enhance production and create employment for her citizen.

iii. High Tariff

The interest rate placed on the purchase of a brand new car in the country today is on the high side, the cost of purchasing a new car in Nigeria is very high, such that an average Nigeria cannot afford to buy a car of his choice.

iv. Poor Economy

The poor economy of the nation has hiking the price and maintenance of Automobile cars in Nigeria, owing a car in Nigeria is still a big thing.

v.

Lack Of Fund

The cost of running any industry in Nigeria is sole shouldered by the owners of the companies. The Nigeria government never think of a way out to support and encourage the existing companies, by ensuring or creating a conducive atmosphere for them to operate, but rather creating a way to share in the company's profit, by increasing the company tax yearly as a way of paying for their existence in the country.

vi.

Poor power Supply

The power of any industry is electricity, lack of electricity in established industries result to stagnation

of the company in terms of improvement and development. Most developed countries in the western world are able to survive based on the constant of electricity that is used in the operation of their machines. Automobile industry can only survive when there is a constant supply of electricity.

vii.

THE HEAT ENGINE

Heat engine is a device that converts heat to work. It takes heat from a reservoir, then does some work like moving a piston, lifting weight etc. and finally discharges some heat energy into the sink.

Difference Between Petrol and Diesel Engine	
Diesel Engine	Petrol Engine
These engines work on the Diesel cycle	Works on the Otto cycle
The fuel is mixed with air inside the cylinder	Air and the fuel are mixed in a carburetor
Ignition is achieved with the help of the hot, compressed air.	Fuel is ignited with an electric spark
High compression ratio	Relatively low compression ratio
High power production	Relatively low amounts of power are produced in a Petrol engine
These engines work with fuels that have low volatilities	Highly volatile fuels are used in these internal combustion engines
Generally used in heavy vehicles such as trucks and buses	Used in light vehicles such as motorcycles and cars.
Relatively low fuel consumption	High fuel consumption.
High initial and maintenance costs	Comparatively low initial cost and maintenance cost

INTERNAL AND EXTERNAL COMBUSTION ENGINES

i. Internal combustion engine (IC)

In these heat engines, the fuel burns inside the cylinder. A car engine is an example of an internal combustion engine.

ii. External combustion engine (EC)

In these heat engines, the fuel burns outside and away from the main engine, where force and motion are produced. A steam engine is an example of an external combustion engine.

iii. Valve Timing

In internal combustion engines, there are at least two valves for every cylinder: an intake valve and an exhaust valve. In order for the engine to operate properly, the intake valve must draw in the fuel and air at the proper time, the cylinder fires, and then the exhaust valve must open to allow the burnt fuel to escape so that the cycle can repeat. For the engine to function properly, the precise actuation of these valves, in the proper sequence, is the valve timing

iv. Valve Overlap

Valve overlap is the period during engine operation when both intake and exhaust valves are open at the same time. Valve overlap occurs when the piston nears TDC between the exhaust event and the intake event. Duration of valve overlap is between 10° - 20° of crankshaft rotation, depending on the engine design.

COMPONENTS OF IC ENGINE

• Crankshaft • Cylinder Head • Flywheel • Cylinder Block • Carburetor • Piston • Exhaust Valve • Timing Belt Cover • Con-rod • Timing Belt • Fuel Injector • Oil Pump • Catalyst Converter • Oil Seal • Turbo Charger • Fuel Tank • Sensor • Intake Manifold • Water Pump • Exhaust Manifold • Fan • Radiator • Spark Plug

SOME COMPONENTS OF IC ENGINE

1. Cylinder block

Cylinder is the main body of IC engine. Cylinder is a part in which the intake of fuel, compression of fuel and burning of fuel take place. The main function of cylinder is to guide the piston. It is in direct contact with the products of combustion so it must be cooled. For cooling of cylinder, a water jacket (for liquid cooling used in most of cars) or fin (for air cooling used in most of bikes) are situated at the outer side of cylinder. At the upper end of cylinder, cylinder head and at the bottom end crank case is bolted. The upper side of cylinder is consisting a combustion chamber where fuel burns. To handle all this pressure and temperature generated by combustion of fuel, cylinder material should have high compressive strength. So it is made by high grade cast iron. It is made by casting and usually cast in one piece. 2

2. Cylinder head

The top end of the engine cylinder is closed by means of removable cylinder head. There are two holes or ports at the cylinder head, one for intake of fuel and other for exhaust. Both the intake and exhaust ports are closed by the two valves known as inlet and exhaust valve. The inlet valve, exhaust valve, spark plug, injector etc. are bolted on the cylinder head. The main function of cylinder head is to seal the cylinder block and not to permit entry and exit of gases on cover head valve engine. Cylinder head is usually made by cast iron or aluminum. It is made by casting or forging and usually in one piece. 2

3. Connecting rod

Connecting rod connects the piston to crankshaft and transmits the motion and thrust of piston to crankshaft. It converts the reciprocating motion of the piston into rotary motion of crankshaft. There are two end of connecting rod, one is known as big end and other as small end. Big end is connected to the crankshaft and the small end is connected to the

piston by use of piston pin. The connecting rods are made of nickel, chrome, and chrome vanadium steels. For small engines the material may be aluminum. 2

4. Crankshaft

The crankshaft of an internal combustion engine receives the efforts or thrust supplied by piston to the connecting rod and converts the reciprocating motion of piston into rotary motion of crankshaft. The crankshaft mounts in bearing so it can rotate freely. The shape and size of crankshaft depends on the number and arrangement of cylinders. It is usually made by steel forging, but some makers use special types of cast-iron such as spheroidal graphitic or nickel alloy castings which are cheaper to produce and have good service life.

5. Manifold

The main function of manifold is to supply the air fuel mixture and collects the exhaust gases equally from all cylinder. In an internal combustion engine two manifold are used, one for intake and other for exhaust. They are usually made by aluminum alloy. 2

THE CARBURATOR

A carburetor is a device which helps in mixing fuel and air together for facilitating internal combustion inside an internal combustion engine. This device passes the mixture of fuel and air to the intake manifold (a device delivering air/fuel mixture to the cylinders) of an internal combustion engine.

COMPONENTS OF THE CARBURATOR

- Components of carburetors usually include a storage chamber for liquid fuel, a choke, an idling (or slow-running) jet, a main jet, a venturi-shaped air-flow restriction, and an accelerator pump. The quantity of fuel in the storage chamber is controlled by a valve actuated by a float.
- Atomization is the process of converting an analyte in solid, liquid or solution form to a free gaseous atom. It is the transformation of a bulk liquid into a spray of liquid droplets in a surrounding gas or vacuum.
 - Vaporization, conversion of a substance from the liquid or solid phase into the gaseous (vapour) phase. If conditions allow the formation of vapour bubbles within a liquid, the vaporization process is called boiling. Direct conversion from solid to vapour is called sublimation.
 - The process of mixing (as in a carburetor) the vapor of a flammable hydrocarbon (such as gasoline) with air to form an explosive mixture especially for use in an internal-combustion engine.
 - The stoichiometric mixture for a gasoline engine is the ideal ratio of air to fuel that burns all fuel with no excess air. For gasoline fuel, the stoichiometric air-fuel mixture is about 14.7:1 i.e. for every one gram of fuel, 14.7 grams of air are required.
 - A rich mixture is a fuel/air mixture containing an excessive proportion of fuel. A rich mixture may be desirable under certain operating conditions, such as when the engine is under load. A rich mixture provides enough fuel to use up all of the oxygen in the cylinder.

vi. A lean mixture is a fuel/air mixture containing a relatively low proportion of fuel. A lean mixture can be caused by too little fuel or too much air. Too much oxygen indicates a lean mixture and the need for less fuel.

- ii. The maximum power theorem, better known as the maximum power transfer theorem, is an essential tool for ensuring successful system design. Put simply, this theorem states that the maximum power that can be transferred from source to load is 50%, which occurs when source impedance is exactly matched to load impedance.
- iii. Economic ratios convert between various economic measures, e.g., direct spending to the directly associated jobs, labor income, and value added in each sector.

THE SUPERCHARGER

A supercharger is a device which increases the induction pressure of an engine and is connected between the carburetor and the induction manifold.

The purpose of supercharging can be stated as:

1. To reduce the weight per horsepower of the engine.
2. To minimize the space occupied by the engine.
3. To maintain the power of the engine even at high altitudes.
4. To improve power in a racing car.
5. To improve combustion efficiency due to the formation of a homogeneous mixture.

Differences between superchargers and turbo chargers:-

1. Superchargers are powered by the engine itself and use a fraction of crankshaft power, while turbochargers are powered by a turbine driven by the engine exhaust, therefore, do not use any of the crankshaft power
2. Superchargers are less thermal efficient than the turbochargers.
3. Superchargers have a shorter response to the throttle than the turbochargers.

TYPES OF IGNITION SYSTEM

- i. Conventional Breaker-Point Ignition System

The conventional breaker-point ignition system is the oldest type of ignition system and has been used since the early days of the automobile, especially through the 1970s. The mechanical nature of these ignition systems, as well as the length of time these systems

have been used, they are relatively easy to diagnose and repair. However, they do incorporate a large number of moving parts, increasing the potential for breakdowns. Additionally, deterioration of these systems can negatively affect the maximum span

energy throughout the life of the engine, causing frequent misfires and increasing emissions.

ii. High Energy (Electronic) Ignition System

After 70-plus years of using conventional breaker-point ignition systems, automotive manufacturers turned to a more advanced, high energy ignition system. This system replaces the breaker points and condenser with a transistorized switch within an ignition module that also handles the task of triggering the ignition coil to generate high-voltage current. This can prove advantageous, as the use of this electronic switch means there are fewer moving parts than in a breaker-point ignition system while still being relatively easy to diagnose and repair. They can also provide a consistent, high voltage spark throughout the life of the engine, meaning fewer misfires.

But while the breaker points have been replaced in a high energy ignition system, these systems still use a distributor cap and rotor to perform the same job of distributing current to the spark plugs. Using a conventional distributor means they will eventually wear and require replacing, adding to potential repair costs. Also, the ignition timing cannot be as precisely controlled as with more sophisticated systems, resulting in poor fuel efficiency and lethargic acceleration.

iii. Distributor-less (Waste Spark) Ignition System

Knowing the potential issues associated with a conventional distributor, this aptly named system eliminates the distributor entirely. Instead, multiple ignition coils are used—one for each pair of cylinders. Using engine sensors to determine crankshaft position and camshaft position, an Electronic Control Unit triggers the appropriate ignition coil and directs the distribution of electrical current to the spark plugs. This system also uses a “Waste Spark” for one of the paired cylinders, pairing two pistons that will be at the top dead center at the same time—one at the end of its compression stroke, and the other at the end of its exhaust stroke. Each of the spark plugs in these cylinders will fire at the same time using the high voltage from one coil. This is beneficial, as this system can be designed to generate high voltage, and ignition timing can be precisely controlled for lower emissions. In addition, the lack of moving parts means lower maintenance costs. Unfortunately, this also means that it can be much harder to diagnose and is more expensive than a traditional system. This system also requires double platinum to reverse firing.

iv. Coil-on-Plug (Direct) Ignition System

The most sophisticated of all ignitions systems, this system places an ignition coil directly on the top of each spark plug and is ideal for modern engines. All of the ignition

timing is handled by the Engine Control Unit, based on input from various sensors. Because each spark plug has its own dedicated coil, high-voltage spark plug wires are completely eliminated. This is beneficial for a number of reasons: there are no moving parts and lower maintenance costs. It should be noted that this can make it more difficult to diagnose and more expensive to repair than a traditional system, but with lower maintenance costs, repairs are less frequent. Through the use of the Engine Control Unit, this system can be designed to generate high voltage, and the ignition timing can be **precisely controlled for lower emissions.**

FUNCTION OF THE IGNITION SYSTEM

The ignition system's job is to **ignite the air-fuel mixture in a gasoline engine.** In addition to the spark plugs that spark in the combustion chambers, the ignition system consists of an ignition coil, which converts battery voltage to higher voltage, and the spark plug wires that join these components.

The fuel injection types used in newer cars include four basic types:

- Single-point or throttle body injection.
- Port or multipoint fuel injection.
- Sequential fuel injection.
- Direct injection.

Functions following ignition system components:

- (i) Ignition coil: Ignition coil receives voltage from primary circuit and convert it to very high voltage up to 28000 volts required for spark by means of primary and secondary winding.
- (ii) Distributor: Distributor rotor rotates and supplies this high voltage current to proper spark plug depending upon the engine firing order.
- (iii) Condenser: The Function of the condenser is to reduce arcing at the contact breaker points. When the magnetic field is collapsing condenser gets fully charged and then it starts discharging and helps in building up of voltage in secondary winding.
- (iv) Spark plug: In S.I engine at the end of compression stroke Charge of air and petrol is ignited by means of the spark produced by spark plug.

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